Supplementary Material for Multi-Label Meta-Learning for Pipeline Recommendation: The importance of pipeline diversity

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Abstract. This document presents the supplementary material for the paper "Multi-Label Meta-Learning for Pipeline Recommendation: The importance of pipeline diversity", submitted to BRACIS 2026. It contains the details that were not included in the paper. In addition to this document, the codes can be found on Github: https://github.com/cynthiamaia/MetaLearningPipeline.

I. META-FEATURES

Table I shows the eighty-one meta-features used to represent each dataset.

TABLE I: Meta-features used to construct the meta-dataset.

Meta-feature	Description	Group
attr_conc.mean	Concentration coef. of each pair of distinct attributes.	info-theory
attr_conc.sd	Concentration coef. of each pair of distinct attributes.	info-theory
attr_ent.mean	Shannon's entropy for each predictive attribute.	info-theory
attr_ent.sd	Shannon's entropy for each predictive attribute.	info-theory
class_conc.mean	Concentration coefficient between each attribute and class.	info-theory
class_conc.sd	Concentration coefficient between each attribute and class.	info-theory
class_ent	Target attribute Shannon's entropy.	info-theory
eq_num_attr	Number of attributes equivalent for a predictive task.	info-theory
joint_ent.mean	Joint entropy between each attribute and class.	info-theory
joint_ent.sd	Joint entropy between each attribute and class.	info-theory
mut_inf.mean	Mutual information between each attribute and target.	info-theory
mut_inf.sd	Mutual information between each attribute and target.	info-theory
ns_ratio	Noisiness of attributes.	info-theory
leaves	Number of leaf nodes in the DT model.	model-based
leaves_branch.mean	Size of branches in the DT model.	model-based
leaves_branch.sd	Size of branches in the DT model.	model-based
leaves_corrob.mean	Leaves corroboration of the DT model.	model-based
leaves_corrob.sd	Leaves corroboration of the DT model.	model-based
leaves_homo.mean	DT model Homogeneity for every leaf node.	model-based
leaves_homo.sd	DT model Homogeneity for every leaf node.	model-based
leaves_per_class.mean	Proportion of leaves per class in DT model.	model-based
leaves_per_class.sd	Proportion of leaves per class in DT model.	model-based

Meta-feature	Description	Group
nodes	Number of non-leaf nodes in DT model.	model-based
nodes_per_attr	Ratio of nodes per number of attributes in DT model.	model-based
nodes_per_inst	Ratio of non-leaf nodes per number of instances in DT model.	model-based
nodes_per_level.mean	Ratio of number of nodes per tree level in DT model.	model-based
nodes_per_level.sd	Ratio of number of nodes per tree level in DT model.	model-based
nodes_repeated.mean	Number of repeated nodes in DT model.	model-based
nodes_repeated.sd	Number of repeated nodes in DT model.	model-based
tree_depth.mean	Depth of every node in the DT model.	model-based
tree_depth.sd	Depth of every node in the DT model.	model-based
tree_imbalance.mean	Tree imbalance for each leaf node.	model-based
tree_imbalance.sd	Tree imbalance for each leaf node.	model-based
tree_shape.mean	Tree shape for every leaf node.	model-based
tree_shape.sd	Tree shape for every leaf node.	model-based
var_importance.mean	Features importance of the DT model for each attribute.	model-based
var_importance.sd	Features importance of the DT model for each attribute.	model-based
cov.mean	Absolute value of the covariance of distinct dataset attribute pairs.	statistical
cov.sd	Absolute value of the covariance of distinct dataset attribute pairs.	statistical
eigenvalues.mean	Eigenvalues of covariance matrix from dataset.	statistical
eigenvalues.sd	Eigenvalues of covariance matrix from dataset.	statistical
gravity	Distance between minority and majority classes center of mass.	statistical
iq_range.mean	Interquartile range (IQR) of each attribute.	statistical
iq_range.sd	Interquartile range (IQR) of each attribute.	statistical
mad.mean	Median Absolute Deviation (MAD) adjusted by a factor.	statistical
mad.sd	Median Absolute Deviation (MAD) adjusted by a factor.	statistical
nr_cor_attr	Number of distinct highly correlated pair of attributes.	statistical
nr_norm	Number of attributes normally distributed based in a given method.	statistical
nr_outliers	Number of attributes with at least one outlier value.	statistical
sparsity.mean	Sparsity metric for each attribute.	statistical
sparsity.sd	Sparsity metric for each attribute.	statistical
t_mean.mean	Trimmed mean of each attribute.	statistical
t_mean.sd	Trimmed mean of each attribute.	statistical
attr_to_inst	Ratio between the number of attributes.	simple
cat_to_num	Ratio between the number of categoric and numeric features.	simple
freq_class.mean	Relative frequency of each distinct class.	simple
freq_class.sd	Relative frequency of each distinct class.	simple
inst_to_attr	Ratio between the number of instances and attributes.	simple
nr_attr	Total number of attributes.	simple
nr_bin	Number of binary attributes.	simple
nr_cat	Number of categorical attributes.	simple
nr_class	Number of distinct classes.	simple

Meta-feature	Description	Group
nr_inst	Number of instances (rows) in the dataset.	simple
nr_num	Number of numeric features.	simple
c1	Entropy of class proportions.	complexity
c2	Imbalance ratio.	complexity
cls_coef	Clustering coefficient.	complexity
density	Average density of the network.	complexity
f1v.mean	Maximum Fisher's discriminant ratio.	complexity
f3.mean	Feature maximum individual efficiency.	complexity
f4.mean	Collective feature efficiency.	complexity
l1.mean	Sum of error distance by linear programming.	complexity
l2.mean	OVO subsets error rate of linear classifier.	complexity
l3.mean	Non-Linearity of a linear classifier.	complexity
n3.mean	Mean error rate of the nearest neighbor classifier.	complexity
n3.sd	Error rate of the nearest neighbor classifier.	complexity
n4.mean	Non-linearity of the k-NN Classifier.	complexity
n4.sd	Compute the non-linearity of the k-NN Classifier.	complexity
t2	Average number of features per dimension.	complexity
t3	Average number of PCA dimensions per points.	complexity
t4	Ratio of the PCA dimension to the original dimension.	complexity

The list of the 49 meta-features excluded for having more than 10% missing values is presented in Table II

TABLE II: Meta-features excluded.

Meta-feature	Description	Group
range.mean	Range (max - min) of each attribute.	statistical
range.sd	Range (max - min) of each attribute.	statistical
mean.mean	Mean value of each attribute.	statistical
mean.sd	Mean value of each attribute.	statistical
min.mean	Minimum value from each attribute.	statistical
min.sd	Minimum value from each attribute.	statistical
roy_root	Roy's largest root.	statistical
median.mean	Median value from each attribute.	statistical
median.sd	Median value from each attribute.	statistical
nr_disc	Number of canonical correlation between each attribute and class.	statistical
max.sd	Maximum value from each attribute.	statistical
max.mean	Maximum value from each attribute.	statistical
w_lambda	Wilks' Lambda value.	statistical
lh_trace	Lawley-Hotelling trace.	statistical
sd.mean	Standard deviation of each attribute.	statistical
can_cor.sd	Canonical correlations of data.	statistical
can_cor.mean	Canonical correlations of data.	statistical
sd.sd	Standard deviation of each attribute.	statistical
sd_ratio	Statistical test for homogeneity of covariances.	statistical
var.mean	Variance of each attribute.	statistical
var.sd	Variance of each attribute.	statistical
p_trace	Pillai's trace.	statistical
g_mean.mean	Geometric mean of each attribute.	statistical
g_mean.sd	Geometric mean of each attribute.	statistical
h_mean.sd	Harmonic mean of each attribute.	statistical
h_mean.mean	Harmonic mean of each attribute.	statistical

Meta-feature	**Description**	**Group**
cor.mean	Absolute value of the correlation of distinct dataset column pairs.	statistical
cor.sd	Absolute value of the correlation of distinct dataset column pairs.	statistical
kurtosis.sd	kurtosis of each attribute.	statistical
kurtosis.mean	kurtosis of each attribute.	statistical
skewness.sd	Skewness for each attribute.	statistical
skewness.mean	Skewness for each attribute.	statistical
num_to_cat	Number of numerical and categorical features.	simple
f1v.sd	Maximum Fisher's discriminant ratio.	complexity
l3.sd	Non-Linearity of a linear classifier.	complexity
l1.sd	Sum of error distance by linear programming.	complexity
l2.sd	OVO subsets error rate of linear classifier.	complexity
f4.sd	Collective feature efficiency.	complexity
f3.sd	Feature maximum individual efficiency.	complexity
f2.mean	Volume of the overlapping region.	complexity
n2.sd	Ratio of intra and extra class nearest neighbor distance.	complexity
n2.mean	Ratio of intra and extra class nearest neighbor distance.	complexity
f1.sd	Maximum Fisher's discriminant ratio.	complexity
f1.mean	Maximum Fisher's discriminant ratio.	complexity
f2.sd	Volume of the overlapping region.	complexity
n1	Fraction of borderline points.	complexity
lsc	Local set average cardinality.	complexity
hubs.sd	Hub score.	complexity
hubs.mean	Hub score.	complexity